

# Prompt and Format Robustness in LLM Evaluation

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# Motivation

- LLM evaluations often report one score from one prompt template.
- Prior work shows that seemingly minor prompt-format choices can move measured performance.
- This creates an evaluation problem: a reported score may partly reflect prompt choice, not just model ability.

## Core idea

Hold the task examples and model fixed, vary only the prompt wrapper and output-format constraint, then measure how much the score changes.

| Paper                          | Role in this project                                                                                                                        |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Sclar et al.<br>Mizrahi et al. | Prompt formatting can cause large measured score changes.<br>Single-prompt evaluation is unreliable; multi-prompt evaluation is preferable. |
| Hua et al.<br>Zheng et al.     | Some prompt sensitivity may be an artifact of brittle evaluation.<br>LLM-as-judge can be useful as a secondary evaluator or audit.          |

## Project positioning

A small modern retest of prompt sensitivity, plus a scoring-method ablation.

# Research Question

## Main question

How much do semantically equivalent prompt and format changes affect measured LLM performance on structured tasks, and how much does that effect depend on the scoring method?

- Distinguish prompt-associated model behavior from scoring brittleness.
- Use multiple controlled prompt variants instead of one arbitrary template.
- Score the same raw outputs with strict and equivalence-aware evaluators.

$$\begin{aligned} & \mathbf{2 \text{ tasks} \times 100 \text{ examples} \times 8 \text{ prompts} \times 2 \text{ models}} \\ & \mathbf{= 3,200 \text{ model outputs}} \end{aligned}$$

- Task examples are frozen and reused across all prompts and models.
- The dataset question itself is never paraphrased.
- Only the surrounding instruction and requested output format change.

# Tasks and Models

## Tasks

- **TREC-6**: six-way question classification.
- **GSM8K**: math word problems with final numeric answers.

## Models

- gpt-5-nano
- gpt-5-mini

## Model-selection rationale

Use modern API models that are competent enough to avoid random-failure behavior, but not so saturated that ceiling effects hide prompt sensitivity.

**2 wordings  $\times$  4 output formats = 8 prompt variants**

## Instruction wording

- direct
- formal

## Output format

- answer only
- sentence
- tagged: `<answer>...</answer>`
- JSON

## Controlled variation

The prompt variants are intended to be semantically equivalent, but not textually identical.

## Strict scoring

- exact requested format
- exact gold answer
- intentionally brittle

## Equivalence-aware scoring

- extract final answer
- normalize harmless variants
- task-specific and deterministic

### Example

Gold: 5600; model output: 5,600.

Strict: wrong. Equivalence-aware: correct.



## Prompt sensitivity

$$\max_{\text{prompt}} \text{accuracy} - \min_{\text{prompt}} \text{accuracy}$$

## Artifact gap

$$\text{strict sensitivity} - \text{equivalence-aware sensitivity}$$

- Sensitivity measures how much accuracy moves across the 8 prompts.
- Artifact gap measures how much of that movement disappears when harmless formatting differences are normalized.

# Main Result: Prompt Sensitivity

| Task   | Model      | Strict | Equiv. | Artifact Gap |
|--------|------------|--------|--------|--------------|
| GSM8K  | gpt-5-mini | 0.03   | 0.02   | 0.01         |
| GSM8K  | gpt-5-nano | 0.07   | 0.06   | 0.01         |
| TREC-6 | gpt-5-mini | 0.05   | 0.05   | 0.00         |
| TREC-6 | gpt-5-nano | 0.08   | 0.08   | 0.00         |

- Strict prompt sensitivity ranges from 3 to 8 percentage points.
- Equivalence-aware scoring reduces sensitivity slightly on GSM8K.
- TREC-6 sensitivity is unchanged by equivalence-aware scoring.

# Model Quality vs. Robustness

| Task   | Model      | Scorer | Avg. Acc. | Sens. |
|--------|------------|--------|-----------|-------|
| GSM8K  | gpt-5-mini | Strict | 91.1      | 3.0   |
| GSM8K  | gpt-5-nano | Strict | 92.9      | 7.0   |
| TREC-6 | gpt-5-mini | Strict | 88.0      | 5.0   |
| TREC-6 | gpt-5-nano | Strict | 87.2      | 8.0   |

## Interpretation

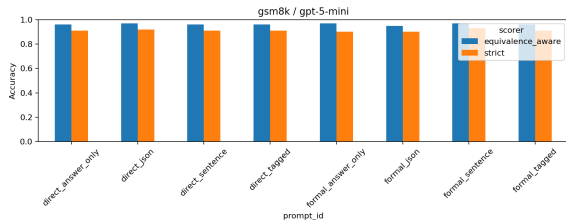
The models are not dramatically separated by average accuracy, but gpt-5-nano is more prompt-sensitive. The interesting difference is robustness, not just raw accuracy.

# Prompt Variant Trends

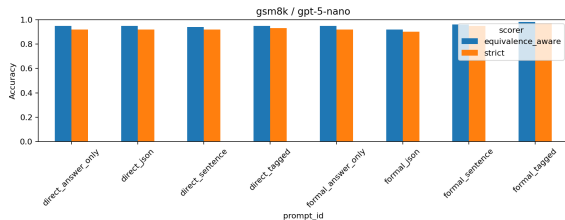
| Prompt Variant     | Strict Avg. | Equiv. Avg. | Delta |
|--------------------|-------------|-------------|-------|
| direct_sentence    | 88.2        | 90.0        | +1.8  |
| direct_answer_only | 89.0        | 91.0        | +2.0  |
| formal_answer_only | 89.0        | 91.5        | +2.5  |
| formal_json        | 89.0        | 90.8        | +1.8  |
| direct_json        | 89.5        | 91.5        | +2.0  |
| direct_tagged      | 90.2        | 92.0        | +1.8  |
| formal_sentence    | 91.2        | 92.5        | +1.3  |
| formal_tagged      | 92.2        | 93.8        | +1.6  |

- `formal_tagged` is strongest on average.
- `direct_sentence` is weakest on average.
- This is descriptive, not a universal prompt recommendation.

# Prompt-Level Accuracy: GSM8K



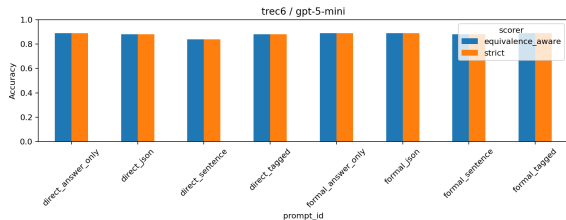
gpt-5-mini



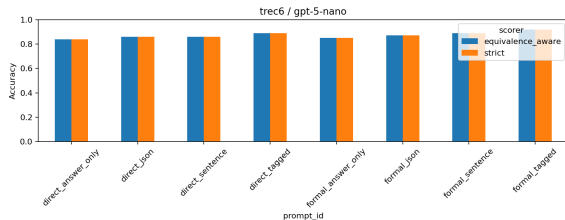
gpt-5-nano

- Equivalence-aware scoring recovers some strict-scoring misses.
- Recovery comes mostly from benign numeric formatting differences.

# Prompt-Level Accuracy: TREC-6



gpt-5-mini



gpt-5-nano

- Strict and equivalence-aware bars overlap.
- The remaining errors are label errors, not formatting artifacts.

## GSM8K

- wrong final numeric values
- correct values with harmless formatting differences
- examples: 5 vs. 5.00, 5600 vs. 5,600

## TREC-6

- wrong category predictions
- common confusions: ENTY vs. DESC, LOC vs. ENTY
- little for equivalence-aware scoring to recover

# LLM-Judge Audit

- Audited all outputs with `strict_correct = false`.
- Judge model: `gpt-5.4`.
- Total judged records: 326.

| Outcome                             | Value |
|-------------------------------------|-------|
| Parse failures                      | 0     |
| Agreement with deterministic scorer | 100%  |
| Judge recovered rule misses         | 0     |
| Judge rejected rule recoveries      | 0     |

## Interpretation

The audit supports that the deterministic equivalence-aware scorer is adequate for these structured tasks.



# Limitations

- Two tasks and two main models.
- One provider.
- 100 examples per task.
- One run per model/example/prompt condition.
- No human evaluation.
- Equivalence-aware scorer is task-specific.

## Interpretation caveat

Small differences of a few percentage points may include finite-sample noise. The result is best read as observed prompt-associated variation, not definitive statistical separation between every prompt template.

# Takeaways

- ① Prompt choice changes measured accuracy even when task, examples, model, and scorer are fixed.
- ② Equivalence-aware scoring removes some formatting artifacts on GSM8K, but prompt sensitivity does not disappear.
- ③ TREC-6 variation is driven by label prediction differences, not scoring artifacts.
- ④ `gpt-5-nano` is more prompt-sensitive than `gpt-5-mini` despite similar average accuracy in some settings.
- ⑤ Multi-prompt evaluation and careful scorer design both matter.

# One-Sentence Summary

## Final thesis

Modern LLM scores can move by several percentage points under semantically equivalent prompt and output-format changes; equivalence-aware scoring removes some formatting artifacts, but prompt sensitivity remains.